

Applied Deep Learning

Omri Azencot

The Stein Faculty of Computer and Information Science
Ben-Gurion University of the Negev

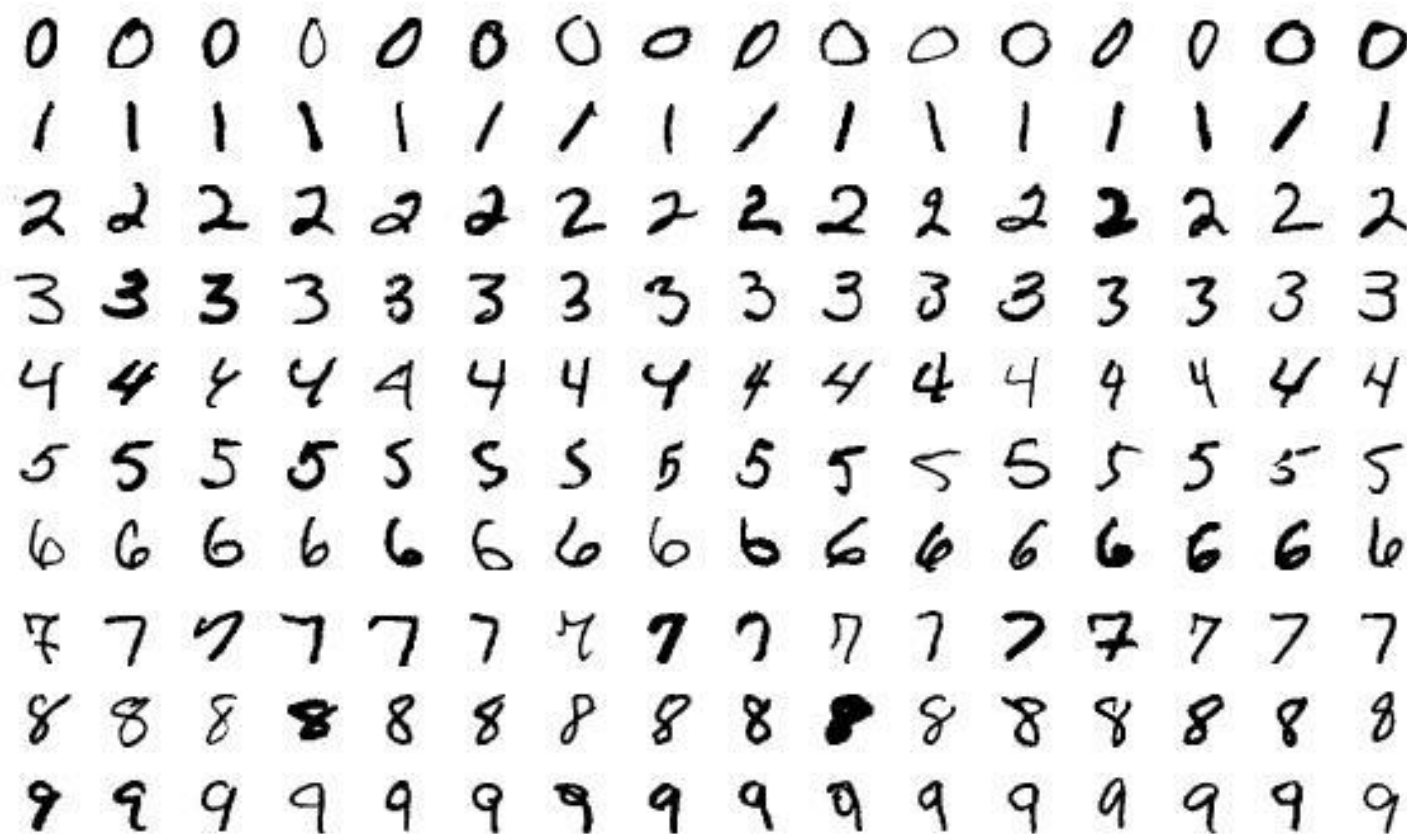
Convolutional Neural Networks



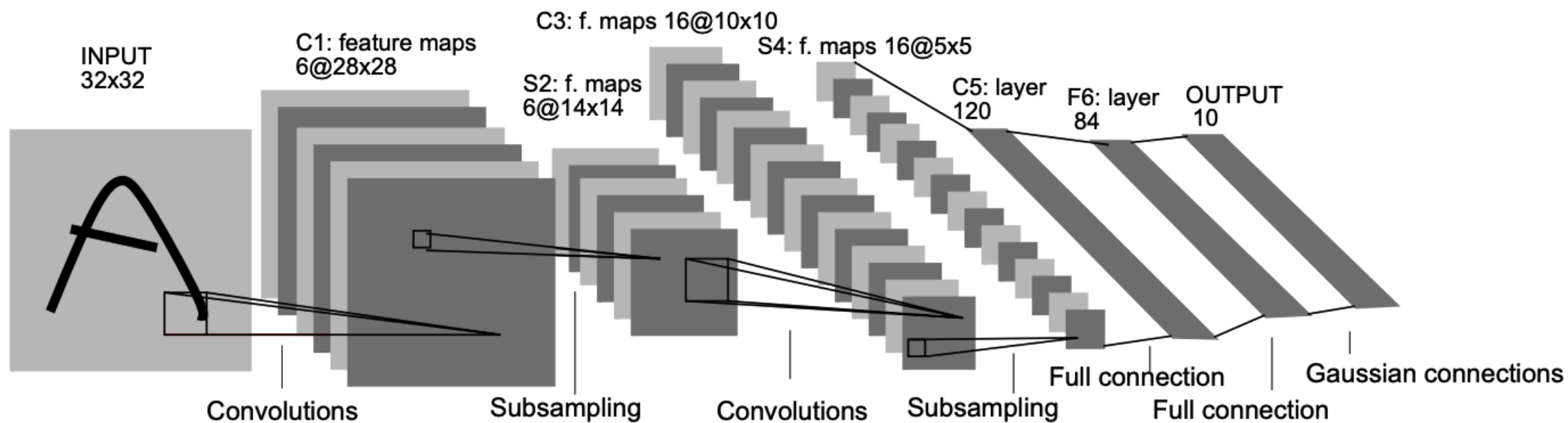
Document Recognition

handwritten numbers

MNIST



LeNet5



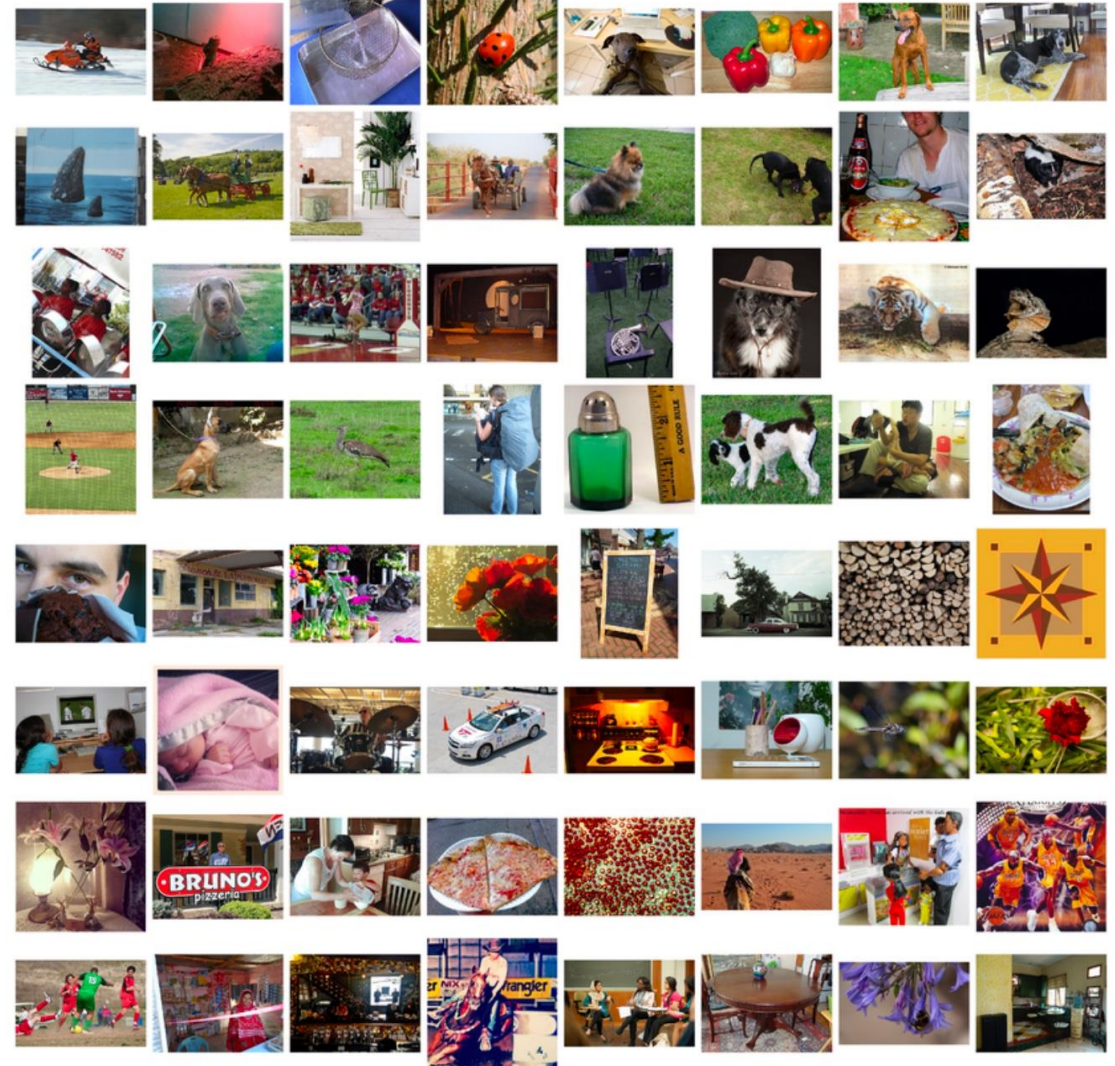
ILSVRC—2012 Competition

Estimate the contents of photos

Train: 1.2 million images
from 1000 categories

validate: 50k images

Test: 100k images



ILSVRC—2012 Competition

Task

For each image, produce 5 object classes l_j in decreasing confidence

$[1, 4, 2, 0, 3]$  $\begin{bmatrix} [0, 1, 0, 0, 0] \\ [0, 0, 0, 0, 1] \\ [0, 0, 1, 0, 0] \\ [1, 0, 0, 0, 0] \\ [0, 0, 0, 1, 0] \end{bmatrix}$

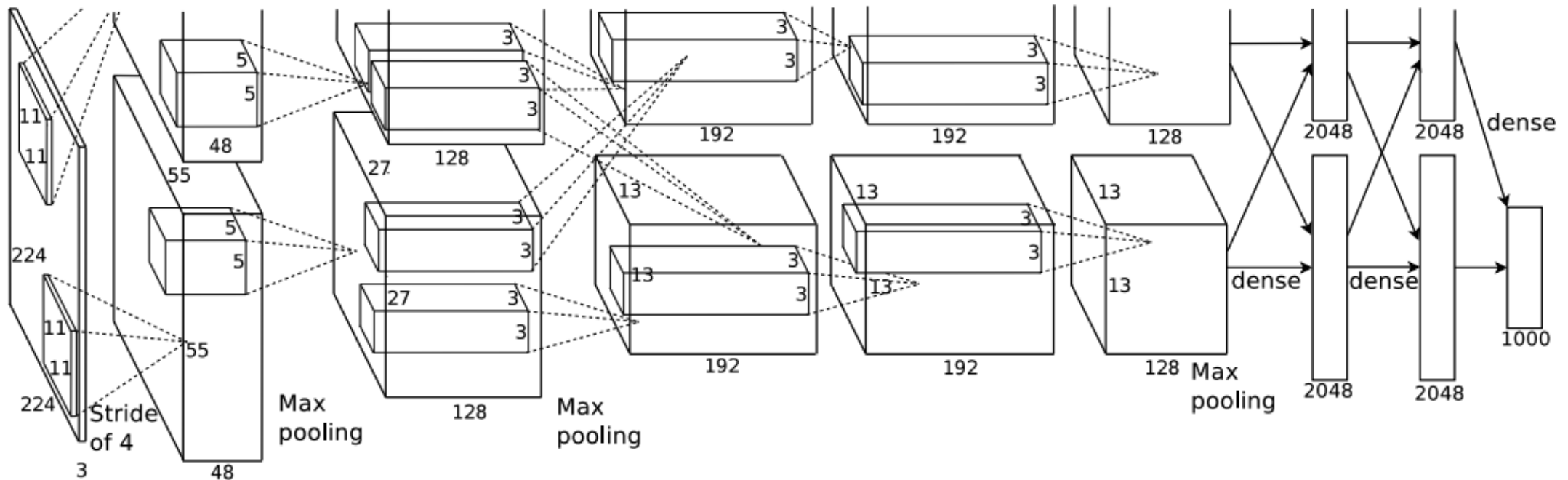
Ground-truth labels $g_k, k = 1, \dots, n$

$$\begin{pmatrix} 0 \\ 1 \\ \dots \\ 0 \end{pmatrix} g_k$$

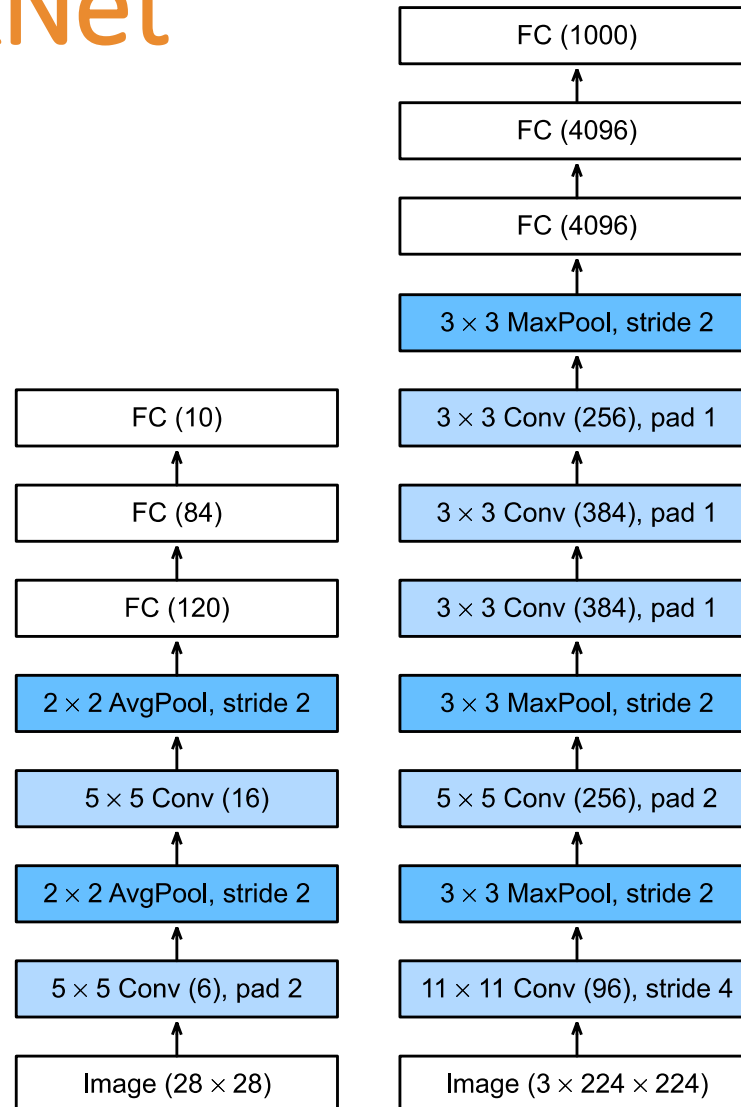
Error

$$e = \frac{1}{n} \cdot \sum_k \min_j d(l_j, g_k)$$

AlexNet



LeNet5 vs. AlexNet

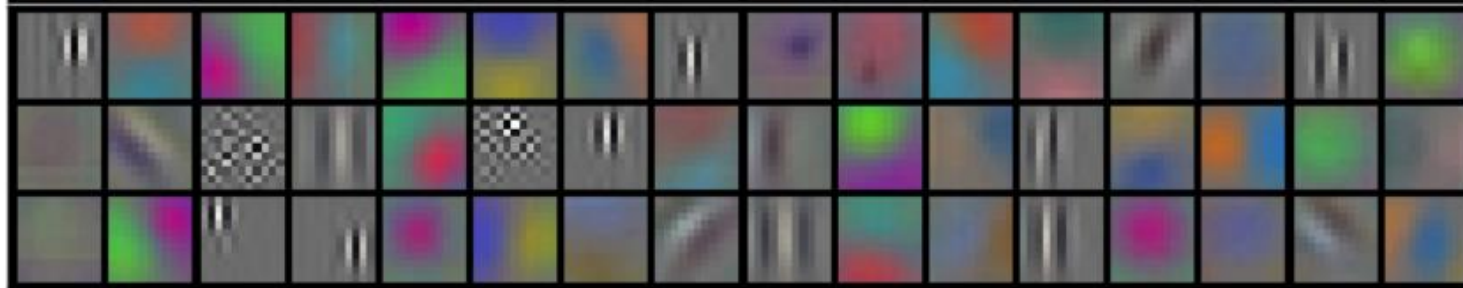


AlexNet Features

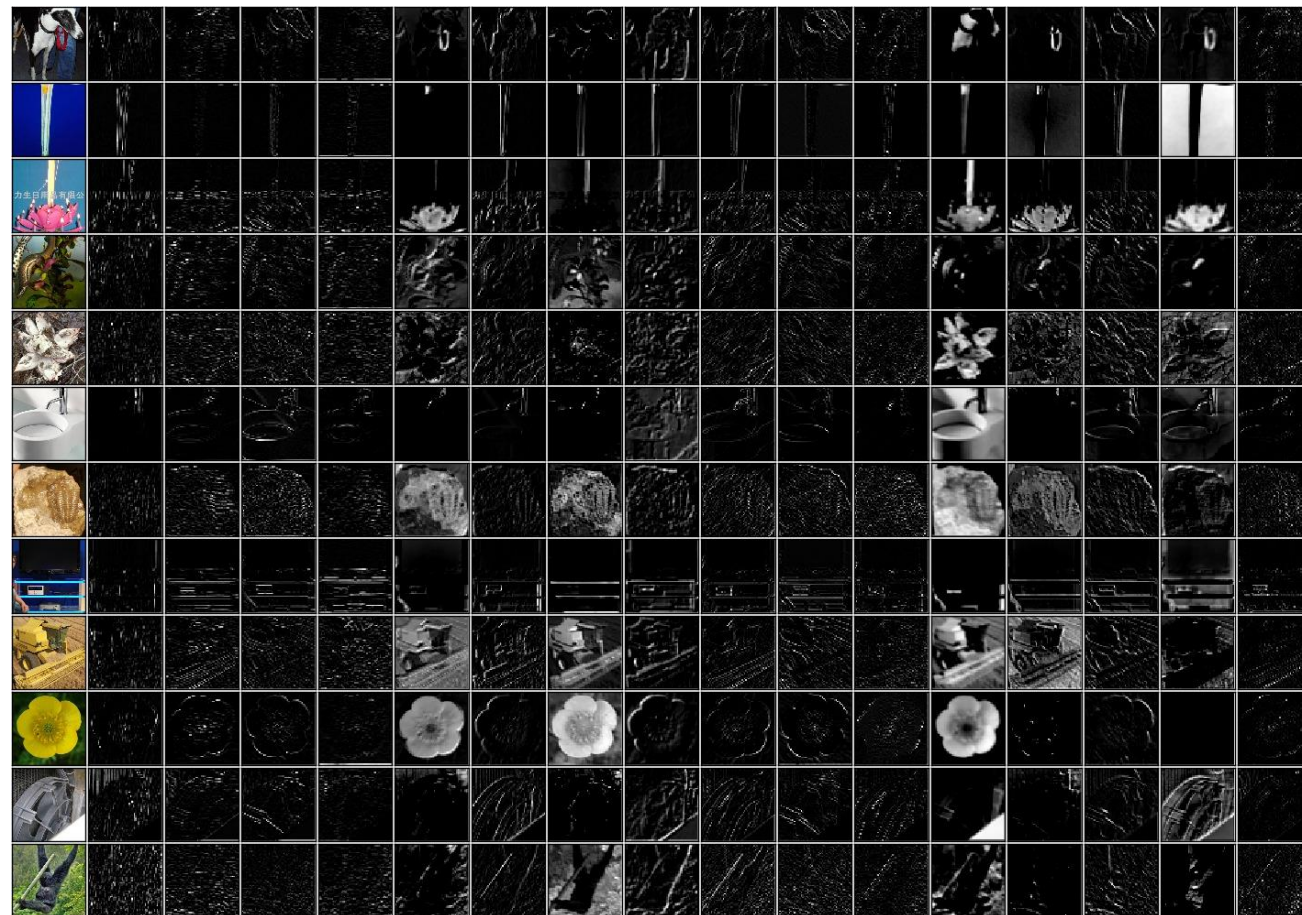
GPU1



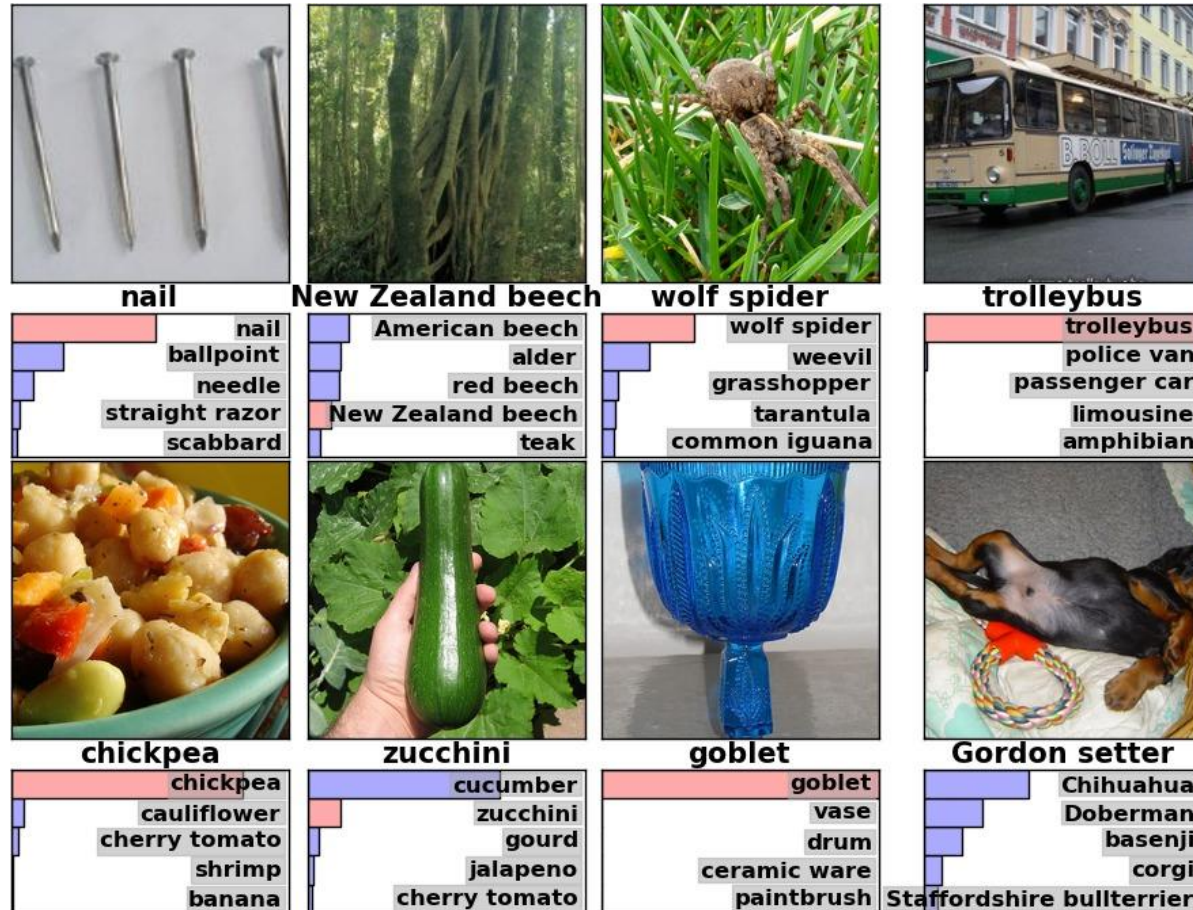
GPU2



AlexNet Features



AlexNet Predictions



VGG Network: Motivation

A convolutional layer



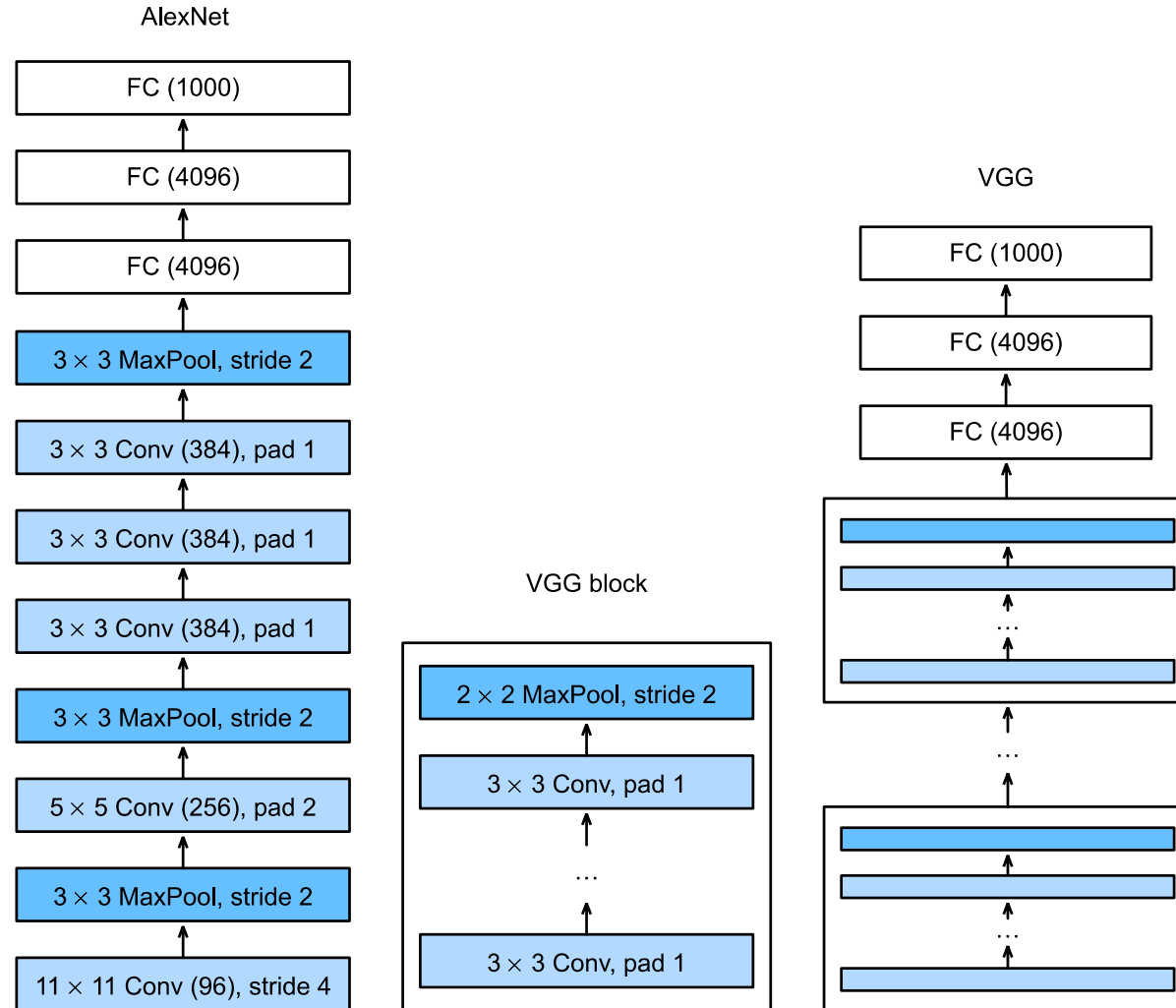
spatial resolution decreases rapidly

$\log_2 d$ convolutional layers

VGG Architecture

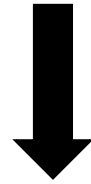
ConvNet Configuration					
A	A-LRN	B	C	D	E
11 weight layers	11 weight layers	13 weight layers	16 weight layers	16 weight layers	19 weight layers
input (224 × 224 RGB image)					
conv3-64	conv3-64 LRN	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64	conv3-64 conv3-64
maxpool					
conv3-128	conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128	conv3-128 conv3-128
maxpool					
conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256	conv3-256 conv3-256 conv1-256	conv3-256 conv3-256 conv3-256	conv3-256 conv3-256 conv3-256 conv3-256
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512	conv3-512 conv3-512 conv1-512	conv3-512 conv3-512 conv3-512	conv3-512 conv3-512 conv3-512 conv3-512
maxpool					
FC-4096					
FC-4096					
FC-1000					
soft-max					

AlexNet vs. VGG

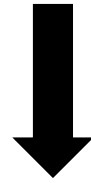


Batch Normalization

deep models convergence is tricky



key idea: maintain a similar scale across layers



helps optimizers: uniform learning rate, regularization, ...

Batch Normalization

Applied per minibatch $\mathcal{B}$

$$\text{BN}(x) = \gamma \odot \frac{x - \hat{\mu}_{\mathcal{B}}}{\hat{\sigma}_{\mathcal{B}}} + \beta$$

mean (points to $\hat{\mu}_{\mathcal{B}}$)

scale (points to γ)

standard deviation (points to $\hat{\sigma}_{\mathcal{B}}$)

shift (points to β)

Mean is computed via

$$\hat{\mu}_{\mathcal{B}} = \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} x$$

Variance is computed via

$$\hat{\sigma}_{\mathcal{B}}^2 = \frac{1}{|\mathcal{B}|} \sum_{x \in \mathcal{B}} (x - \hat{\mu}_{\mathcal{B}})^2 + \epsilon$$

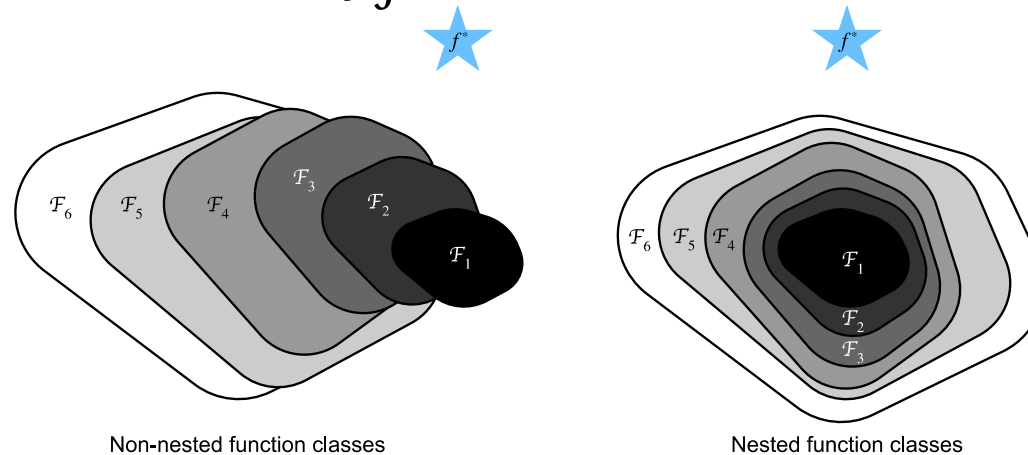


Residual Networks: Motivation

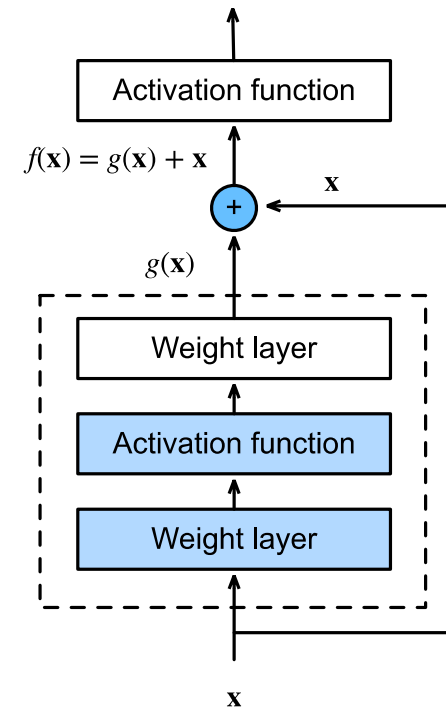
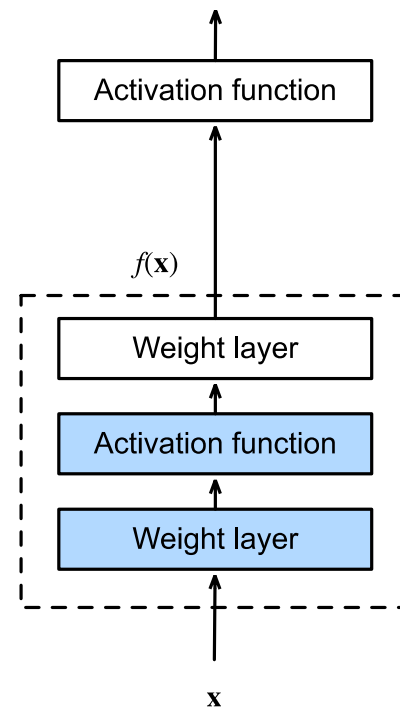
Find f^* using $f \in \mathcal{F}$, namely

$$f_{\mathcal{F}}^* = \arg \min_{f \in \mathcal{F}} L(X, y, f)$$

More powerful $\mathcal{F}'$ yield better $f_{\mathcal{F}'}^*$?



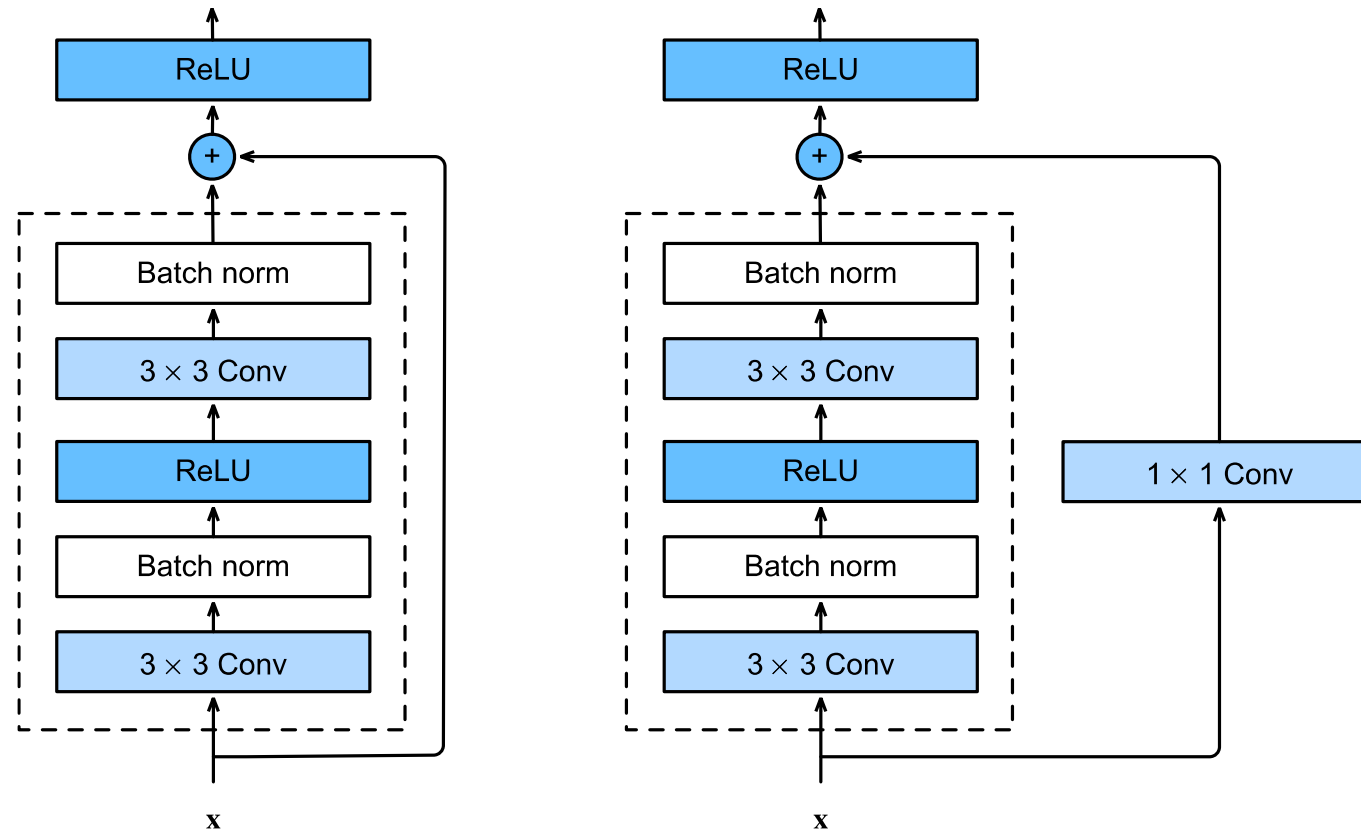
Residual Blocks



*residual
connection*

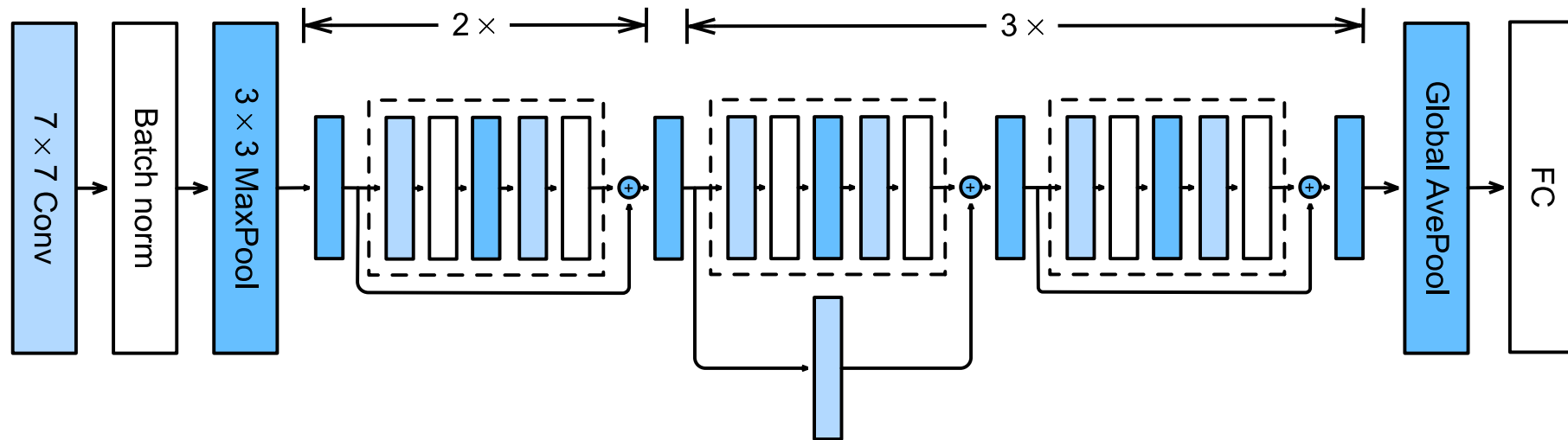


Residual Convolutional Blocks



ResNet Models

ResNet-18



ResNet Models

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
conv2_x	56×56	3×3 max pool, stride 2				
		$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

ResNet Results on ImageNet

single model

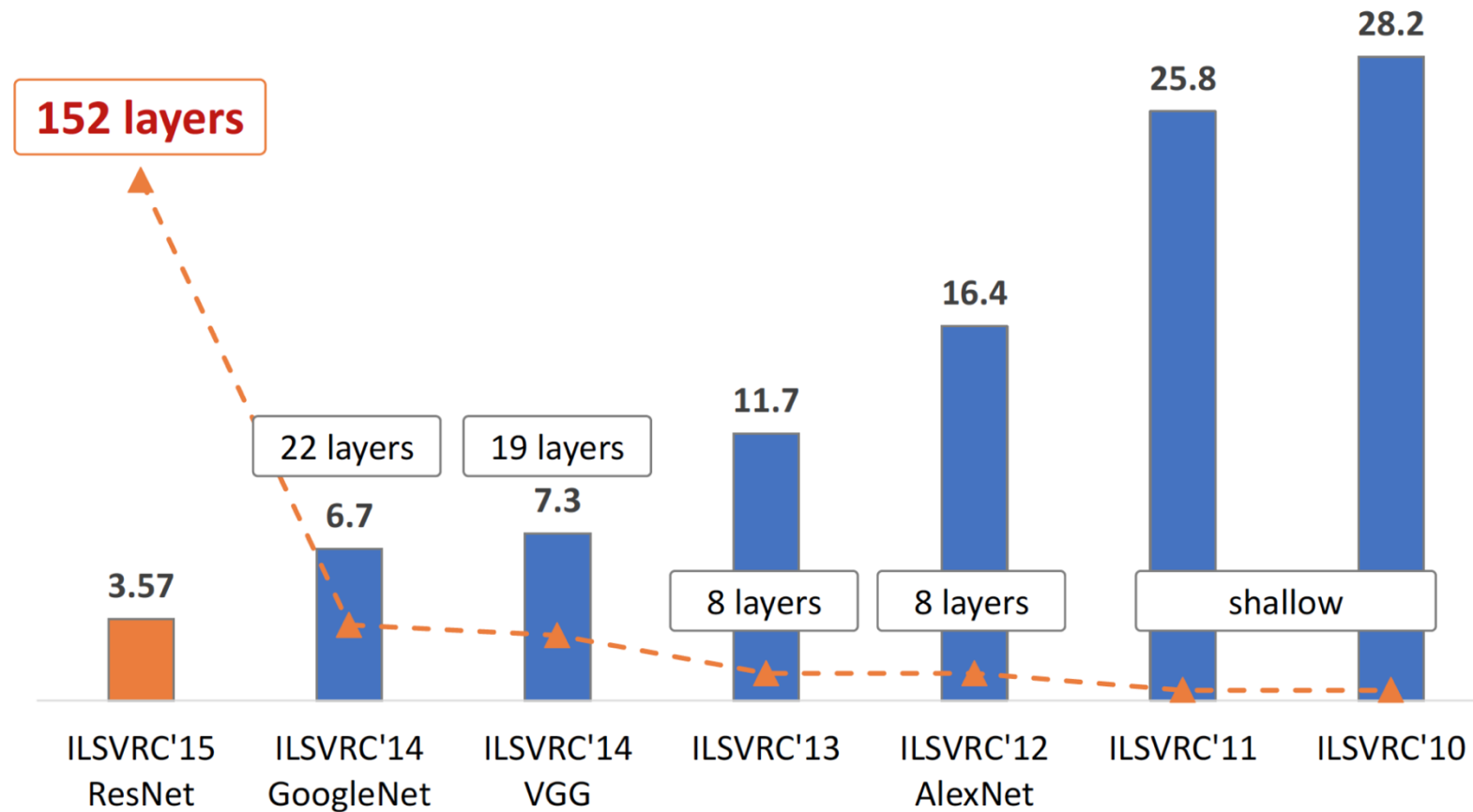
method	top-1 err.	top-5 err.
VGG [41] (ILSVRC'14)	-	8.43 [†]
GoogLeNet [44] (ILSVRC'14)	-	7.89
VGG [41] (v5)	24.4	7.1
PReLU-net [13]	21.59	5.71
BN-inception [16]	21.99	5.81
ResNet-34 B	21.84	5.71
ResNet-34 C	21.53	5.60
ResNet-50	20.74	5.25
ResNet-101	19.87	4.60
ResNet-152	19.38	4.49

ensemble

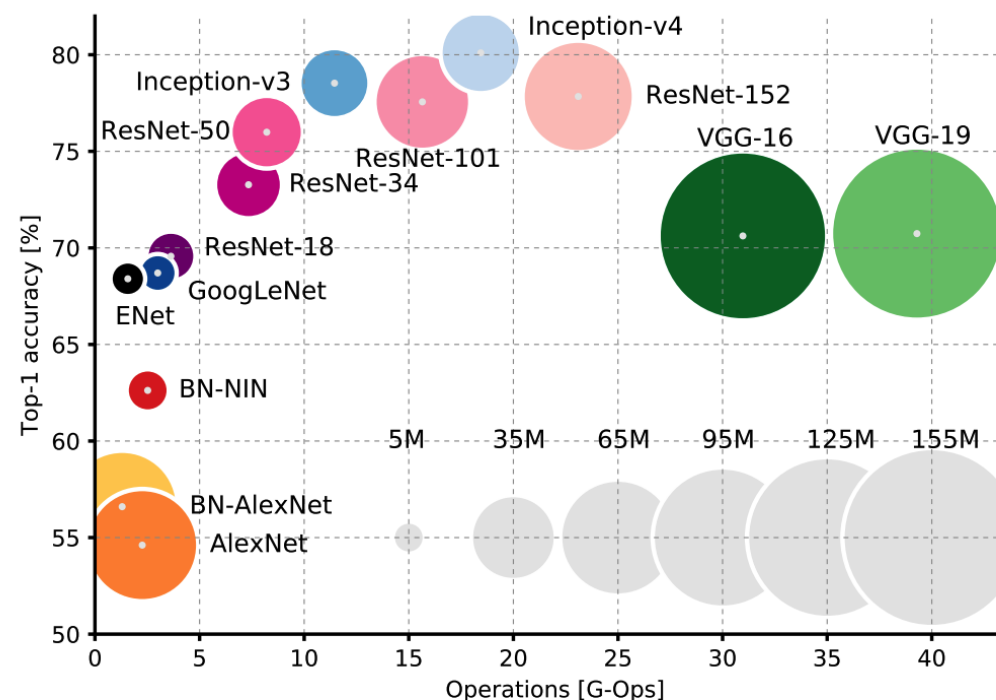
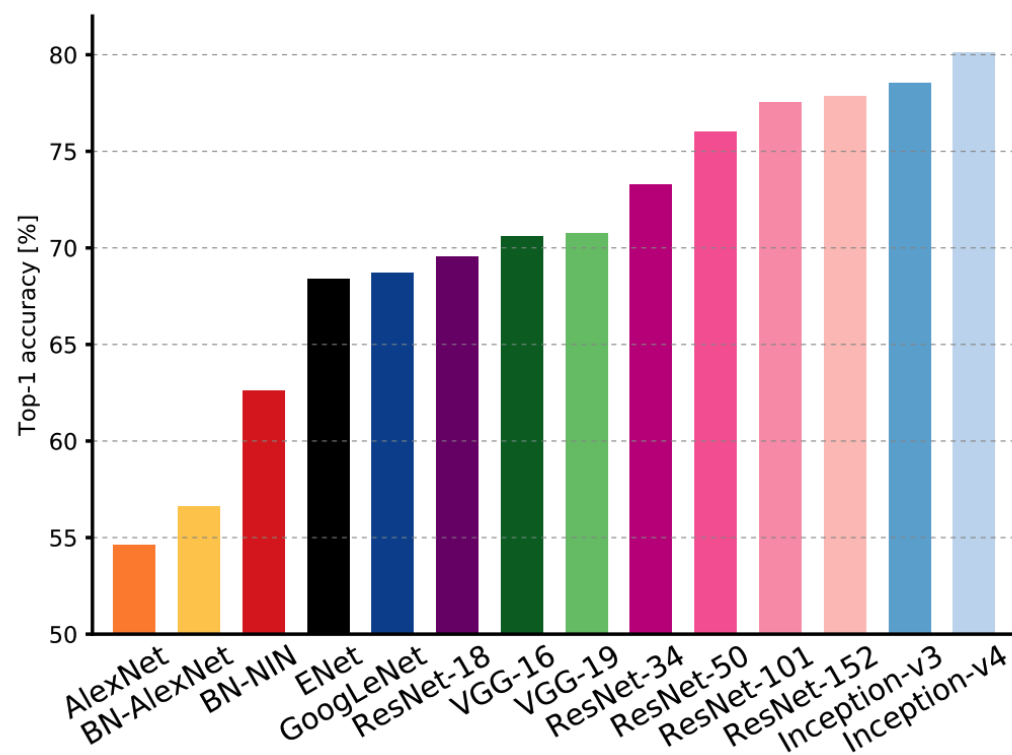
method	top-5 err. (test)
VGG [41] (ILSVRC'14)	7.32
GoogLeNet [44] (ILSVRC'14)	6.66
VGG [41] (v5)	6.8
PReLU-net [13]	4.94
BN-inception [16]	4.82
ResNet (ILSVRC'15)	3.57



Depth Evolution



Error vs. Models/Operations



Code Demo

